Fintech Lending Platform

Cornell Fintech Club — Spring 2026 Project Spec

What We're Building

This project will design and build a full-stack fintech lending platform — the same type of infrastructure that powers companies like Brex, Ramp, and Mercury. The platform covers the complete lending lifecycle: a borrower-facing application portal, an automated credit decisioning engine, and a lender portfolio management dashboard. Optional stretch modules allow members to go deeper into machine learning, LLM/AI tooling, financial modeling, and credit analysis depending on their interests.

Project Structure

The project is divided into a **Core Platform** that all members contribute to, and **Stretch Modules** that members self-select based on interest.

\begin{verbatim}
CORE PLATFORM (all members)
├── Borrower Portal → SWE (frontend) + BA (UX requirements)
├── Rules Engine → SWE (backend) + BA (decision logic)
└── Lender Dashboard → SWE (data viz) + BA (financial metrics)
STRETCH MODULES (self-select)
├── ML Credit Scoring → SWE track
├── LLM Document Parser → SWE track
├── Underwriter AI Chat → SWE track
├── Financial Analysis Model → BA track
├── Credit Memo Generator → BA track

└── Portfolio Risk Dashboard → BA track
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Core Platform Scope

Module 1: Borrower Portal

A clean, user-facing application where a business borrower submits a loan request.
Features:

- Business profile intake (company name, industry, age, structure)
- Loan request details (amount, purpose, term)
- Financial data entry (revenue, EBITDA, existing debt, assets)
- Document upload (financial statements, tax returns, collateral documents)
- Real-time application status tracking

BA Responsibilities: Define all intake fields, validation rules, and the borrower user journey from landing page through submission confirmation
SWE Responsibilities: Build forms, validation, document upload pipeline, and status tracking UI

Module 2: Credit Decisioning Engine

The back-end logic layer that evaluates applications and produces a recommendation.
Features:

- Configurable rule sets by borrower type (SMB, startup, established company)
- Key financial ratio calculations (debt-to-equity, interest coverage, current ratio, DSCR)
- Industry risk tiering (e.g., SaaS vs. manufacturing vs. hospitality)
- Output: Approve / Decline / Manual Review + rationale summary
- Audit log of all decisions and rule triggers

BA Responsibilities: Define credit policy rules, ratio thresholds by borrower category, risk flags, and decision escalation logic
SWE Responsibilities: Implement the rules engine, scoring algorithm, and audit logging system

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Module 3: Lender Dashboard

An internal tool for underwriters and portfolio managers to manage their lending book.
Features:

- Deal pipeline with status tracking (New → In Review → Approved → Funded → Declined)
- Borrower profile view with financial highlights and decision summary
- Portfolio-level metrics (total exposure, approval rate, avg. loan size)
- Search, filter, and sort across all applications

BA Responsibilities: Define metrics displayed, dashboard layout requirements, and filter/sort logic from the underwriter's perspective
SWE Responsibilities: Build the dashboard interface, data queries, and pipeline state management

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Stretch Modules

SWE Track

ML Credit Scoring
Train a predictive default model using publicly available loan data (LendingClub dataset on Kaggle). The model outputs a probability of default that supplements or replaces the static rules engine score.
Skills: scikit-learn, logistic regression, gradient boosting, AUC-ROC evaluation, feature engineering on financial variables
LLM Document Parser
Build a module that uses an LLM (OpenAI API or open-source equivalent) to parse uploaded financial statements and extract structured data — revenue, EBITDA, debt levels, and risk flags — automatically populating the application fields.
Skills: Prompt engineering, document parsing, RAG basics, OpenAI/Anthropic API integration, structured output handling
Underwriter AI Assistant
A chat interface embedded in the lender dashboard where underwriters can ask natural language questions about a deal ("What's this company's interest coverage ratio? How does leverage

compare to the industry median?"). The assistant retrieves and reasons over live deal data.
Skills: LLM orchestration, tool-use / function calling, context management, API design

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BA Track

Borrower Financial Analysis Framework
Design a structured analysis model (Excel or Python) that processes a borrower's uploaded financials and outputs a standardized assessment — EBITDA margins, leverage, liquidity ratios, and year-over-year trends — benchmarked against industry peers.
Skills: Financial ratio analysis, credit metrics, industry benchmarking, financial statement interpretation
Credit Memo Generator
Design the credit memo template that gets auto-populated from the platform's loan data — the same format used by credit analysts at commercial banks. Includes a structured recommendation section with a DCF or comparable company analysis for larger loan requests to contextualize borrower valuation.
Skills: DCF modeling, comparable company analysis, credit writing, structuring arguments for a credit committee
Portfolio Risk & Scenario Dashboard
BAs own both the requirements and the financial logic behind a portfolio-level risk dashboard — weighted average yield, expected default rates, industry concentration risk, and scenario analysis (base / downside / stress cases).
Skills: Portfolio analytics, scenario modeling, risk frameworks, financial dashboarding

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Team Roles

Business Analysts (4–6 members)

Role	Responsibilities
Lead BA	Owns the PRD, coordinates BA/SWE alignment, manages timeline
Process Analyst (1–2)	Maps borrower and underwriter workflows, identifies automation opportunities
Requirements Analyst (1–2)	Writes user stories, acceptance criteria, data dictionaries
Financial Modeling BA (1)	Leads financial analysis framework and credit memo stretch modules

Software Engineers (3–5 members)

Role	Responsibilities
Backend Engineer (1–2)	Rules engine, database schema, API layer
Frontend Engineer (1–2)	Borrower portal, lender dashboard, UI components
ML/AI Engineer (1, optional)	Predictive scoring model and/or LLM integrations

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Key Deliverables

Business Analysis

1. **Product Requirements Document (PRD)** — Full requirements spec covering all three core modules
2. **Process Maps** — Borrower journey and underwriter workflow (current state → future state)
3. **User Stories** — 40–50 stories across all personas (borrower, underwriter, portfolio manager)
4. **Credit Policy Document** — Defined rules, thresholds, and escalation logic for the decisioning engine
5. **Financial Analysis Model** — Ratio framework and benchmark outputs *(stretch)*
6. **Credit Memo Template** — Populated from live deal data with valuation section *(stretch)*

Software Engineering

1. **Deployed Application** — All three core modules functional and integrated
2. **Rules Engine** — Configurable credit scoring logic with audit trail
3. **API Documentation** — OpenAPI spec for all internal endpoints
4. **ML Model** — Trained default prediction model with evaluation metrics *(stretch)*
5. **LLM Integration** — Document parser or underwriter assistant *(stretch)*
6. **Demo Video** — 5-minute end-to-end product walkthrough

Technical Stack (Proposed)

Layer	Options
Frontend	React or Next.js + Tailwind CSS
Backend	Python (FastAPI) or Node.js (Express)
Database	PostgreSQL
ML	scikit-learn, pandas, Python
LLM	OpenAI API or open-source (Mistral, LLaMA)
Storage	AWS S3 or Supabase (free tier)
Project Mgmt	Notion or Linear
Design	Figma
Version Control	GitHub

Skills Members Will Gain

Business Analysis

- Requirements gathering and PRD writing
- Commercial lending process mapping and workflow design
- Credit policy design and structured decision logic
- Financial ratio analysis and credit metrics
- DCF modeling and comparable company analysis *(stretch)*
- Stakeholder management across multiple user personas

Software Engineering

- Rules engine and workflow state machine implementation
- REST API design and documentation
- Dashboard and data visualization development
- ML model training and evaluation on financial data *(stretch)*
- LLM integration, prompt engineering, and tool-use patterns *(stretch)*

Cross-Functional

- Agile BA/SWE collaboration and sprint delivery
- Fintech product development lifecycle
- Corporate credit and lending domain knowledge transferable to banking, PE, and fintech roles

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Project Timeline

Week	Phase	Key Milestones
1–2	Kickoff	Team roles assigned, stretch modules selected, tools set up
3–4	Discovery	Process maps, stakeholder analysis, credit policy defined
5–6	Requirements	PRD finalized, user stories written, API contracts defined
7–9	Core Build	Borrower portal + rules engine MVP functional
10–11	Integration	Dashboard live, modules connected end-to-end
12	Stretch Modules	ML model / LLM integrations / financial models delivered
13	UAT	User acceptance testing, bug fixes, refinement
14	Demo Day	Final presentation, full demo, retrospective

Commitment Expectations

Business Analysts: 6–8 hrs/week, peak 10–12 hrs during requirements and UAT phases
Software Engineers: 8–10 hrs/week, peak 12–15 hrs during core build and stretch module phases
All members are expected to attend weekly syncs and contribute to at least one core module plus optionally one stretch module.

Why This Project

The commercial and SMB lending market represents a multi-trillion dollar opportunity that fintech companies have been actively disrupting over the last decade. Building this platform provides hands-on experience with the exact infrastructure, workflows, and analytical frameworks used at companies ranging from Goldman Sachs and JPMorgan to Brex, Ramp, and Kabbage — making it directly applicable whether members are pursuing investment banking, corporate banking, fintech product roles, software engineering, or data science.